

RITE OF PASSAGE

HANDBOOK

BUILD 260919.2



A transition is not an effect. It is a journey with a landing. One slider carries the music, six slots travel along it on schedules you set, and the arrival is a separate command that lands on the bar, to the sample. [September 2026](#).

ONE SLIDER, WRITTEN AS A SCORE

An insert for the bar before a drop. Put it on a track or a bus, load up to six effects into six **slots**, and tell each slot **where** along one master slider it should travel from its **A** setting to its **B** setting. Then automate that one slider — **POSITION** — from 0 to 100 % over the build.

The slots ignite one after another as the slider passes their entry points. At the end you fire **ARRIVAL**, a separate command, and every slot lets go on the next bar line, to the sample, with a sub-bass impact under it.

Scrubbing the slider **never fires the drop**. Only ARRIVAL does, which is why you can drag POSITION about while you work without the track detonating.

THE SIGNAL PATH

in → slot 1 → slot 2 → ... → slot 6
 → stereo stage (SPREAD / TURN / MONO GATE)
 → MIX → OUTPUT → BAFX rack → out

Slots are in series, top to bottom. **Slot order is chain order.**



Six lanes, each with its own span along the travel. The marker is where POSITION is; a lane takes on heat as the marker crosses it.

WHY A NEW SLOT SEEMS TO DO NOTHING

A slot you have just filled cannot move, and that is deliberate. Choosing an effect seeds **both** sides of the slot — A and B — with that effect's own defaults. A equals B, so the slot resolves to the same value at every position, and **ENTER, EXIT, DEPTH and CURVE all do nothing until you change something in the B row.**

This is on purpose: loading an effect never changes your sound until you ask it to. You can fill all six lanes and hear nothing different, then decide what travels.

THE PANEL SAYS SO

A slot that cannot travel has its span **hatched out** on the lane, its **name in amber** instead of ash, and a line under the A/B editor telling you to move a B knob. A lane is 44 mm of panel with a slider on every pixel, so it has to say it without words.

So the first move after choosing an effect is always the same: **click the lane, then change a knob in the bottom row.**



An empty rack. Nothing is loaded, and the plug-in is bit-transparent: **POSITION** moves and the sound does not.

THE A / B EDITOR

Click a lane to select it. The editor below shows that slot's parameters: top row **A** in yellow, bottom row **B** in ember. A parameter whose A equals its B does not move, however far the marker travels.

ONE LANE, TEN DECISIONS

tick box	slot on or off	DEPTH	how far along the A→B route it actually gets by EXIT. 60 % means it is still on its way when the drop lands, which is often the better sound
socket	the effect's own mark, struck into iron. It warms to ember as the lane catches	PLACE	ST / MID / SIDE / L / R — which part of the stereo field the slot works on
EFFECT	which effect the slot holds. EMPTY is none, duplicates are allowed, and BROKILD WORLD FX at the bottom opens a submenu of ten rack modules. Amber means the slot cannot travel	ON ARRIVAL	STOP kills it dead · SPILL stops feeding it but lets the tail ring on over the downbeat · CLEAR silences it and zeroes its buffers, the vacuum
ENTER	the POSITION at which this slot starts moving from A toward B	CURVE	LIN · ACCEL, late then a rush · DECEL, fast then settling · S · PEAK, which reaches B in the middle and returns to A
EXIT	the POSITION at which it reaches B. Before ENTER it sits at A; after EXIT it holds at B		

THE HOST PARAMETERS

Only these are automatable. Everything else lives in the plug-in state — the rite — on purpose, so re-assigning a slot never orphans an automation lane.

POSITION	0–100 %. The master slider. Automate this
ARRIVAL	a trigger. On its rising edge it arms and fires at the next bar. Set it back to off before the next build
MIX	global dry/wet across the chain
OUTPUT	–24 to +12 dB trim
SPREAD	left and right run the score slightly apart: width without panning
TURN	rotates the field up to ±45°, energy preserving
MONO GATE	how far toward mono the mix collapses over the last 15 % of the travel. ARRIVAL releases it
MACRO 1–5	the World rack's five macros

LET THE PLUG-IN DRIVE THE SLIDER

The strip under POSITION runs the sweep for you off the host transport, so you need not draw an automation lane at all.

AUTO	while it is on, the plug-in owns POSITION and the host parameter is ignored. The slider is disabled and becomes a display
BARS	the cycle length: 1, 2, 4, 8 or 16 bars, repeating
START / END	where in the cycle the sweep runs. These are BAR LINES counted the way a DAW counts them, so in an 8-bar cycle the 8th bar is 8.00 to 9.00. They snap to quarter bars
0 to 100 / 100 to 0	which way it runs
then RESET / then HOLD	what happens between the end of the window and the end of the cycle
ARRIVE	fire ARRIVAL when the sweep completes. Off by default: an automatic drop is a bigger thing than an automatic sweep

WHY RESET IS THE DEFAULT

After the window the position drops straight back. Put the window early in a long cycle and **then HOLD** leaves you at 100 % for several bars, which reads exactly like the sweep having run once and stopped. HOLD is there for when you want the arrival to stay landed until the cycle comes round.

MEASURED

An 8-bar cycle, window on the 8th bar.

transport at bar 4	0.00
bar 8.0	0.00
bar 8.5	0.50
bar 9.0	0.99

With **AUTO off nothing changes at all**. The host parameter drives POSITION exactly as it always did, which is checked so that no existing project moves.

THE EFFECTS ONLY WHILE THE SLIDER MOVES

Most of the time you want the plug-in present during the transition and absent either side of it. The MIX GATE row does that without a second automation lane: it scales MIX by where POSITION is.

FADE IN	shuts the effects off while POSITION is at 0
its length	how much of the travel the fade takes. At 0 the switch is instant and the readout says so: silent at exactly 0 %, fully open one step above it. At 20 % it is half open at 10 % and finished by 20 %
FADE OUT	the same at the other end, measured back from 100 %

There is no click risk at a zero-length gate: the rack already smooths MIX with a 10 ms one-pole.

BOTH OFF MEANS EXACTLY OFF

With neither switch engaged, MIX is exactly the number on the knob. The bench checks that as an IEEE-exact multiply by one rather than a tolerance, so no existing project moves by a hair.

MEASURED

20 % fade in, at 10 %	0.50
... from 20 % on	1.00
25 % fade out, at 87.5 %	0.50
... at 100 %	0.00
both on, at 0 / 50 / 100 %	0.00 / 1.00 / 0.00

WHAT EACH SLOT CAN HOLD

Each carries its own parameters, and every one of them has an A and a B.

CLIMB	resonant filter sweep to self-oscillation — the spine of a build	MODE LP/BP/HP • CUTOFF • RESO • DRIVE
TAPE	analogue echo whose head moves: changing TIME bends the pitch, or crossfades	TIME • FEEDBACK • MOTOR • MODE • DRIVE • WOW • MIX
STUTTER	captures a slice and repeats it, so time stops	DIV • DEPTH • DECAY • PITCH • CAPTURE
CHOP	gates the live signal rhythmically, so time continues	RATE • DEPTH • DUTY • SLEW
RISER	a generator: noise, tone or a Shepard climb, mixed in	SOURCE • FREQ • RESO • LEVEL • WIDTH
GAP	the silence before the drop, placeable anywhere on the score	DEPTH • EDGE
GRAIN	the music breaks into particles, then a cloud	SIZE • DENSITY • SCATTER • SPREAD • MIX
BLOOM	a small room growing into an enormous wash; FREEZE at 100 % holds it for ever	SIZE • DECAY • DAMP • MOD • FREEZE • MIX
FREEZE	the harmonic fingerprint held while the rhythm dissolves	BLUR • HOLD • CAPTURE

... AND THE OTHER NINE

REVERSE	each division played back to front over the next one	DIV · DEPTH · EDGE
BRAKE	the whole signal grinding to a halt, or launching out of one	SPEED 0-200 %
DIVE	the source itself bending into the drop	SEMIS -24...+12 · GRAIN
SWIRL	a room that will not hold still: chorused twenty times as deep as BLOOM's	SIZE · DECAY · WARP · RATE · TONE · MIX
MANGLE	four ways to break it. Travelling ENGINE walks warm, ugly, folded, destroyed	ENGINE · DRIVE · BIAS · CHAR · TONE · MIX
SWARM	one voice becomes a crowd: up to eight copies, each at its own offset	VOICES · DETUNE · DEPTH · RATE · SPREAD · MIX
DUST	older than the record: quantise, hold, wobble the transport, add the room	BITS · RATE · WOW · NOISE · TONE · MIX
ORBIT	round the head, and behind it. ANGLE is where on the circle the source sits	ANGLE · SPIN · WIDTH · REAR · SHADE · MIX
CHANT	the track is made to speak. What arrives is the modulator	CARRIER · PITCH · BANDS · FORMANT · RESPONSE · TONE · MIX

ORBIT'S BACK IS MEASURED

Behind the head it is 3.3 dB quieter, 6.6 dB darker at the top and 24 samples late, because the path round a skull is longer than the path to the front of it. All three together sell the circle; a pan pot gives you only the first.

BRAKE ONLY LAUNCHES OUT OF A STOP

Above 100 % the playback head is trying to catch up with the present, and from rest there is nothing to catch up to. Brake first, then let it go: that is the gesture.

BLOOM'S FREEZE IS EXACT

At 100 % the loop gain is exactly 1.0, the modulation stops and the delays round to whole samples, so a frozen wash neither creeps nor decays. The bench holds it flat over sixty seconds.

TEN MODULES YOU CAN PUT INSIDE THE CHAIN

The bottom of every EFFECT menu opens **BROKILD WORLD FX**: ten modules of the World rack, insertable in a slot like any of the eighteen and swept from A to B like any of the eighteen.

TUBE • SWEEP • ENSEMBLE • HARMONIC • GATE • GRIT • STRIP • ECHO • SHIMMER • ROTARY

The rack is already on this plug-in, post-chain, with its five macros automatable — so this is not about **reaching** those modules. It is about **where they sit**. A slot puts one inside the chain, with its own PLACE and its own ON ARRIVAL, and that buys three things the post-chain rack cannot.

Order	a GRIT before CLIMB is a dirty filter; after it, a clean filter into dirt. Same two boxes, different instrument
Field	GATE set to SIDE rolls the room and leaves the lead alone
The landing	an ECHO set to SPILL rings on through the drop while everything around it is CLEARed



The rack's own face, opened from the header: effects on the left, characters on the right, five macros along the foot.

THEY START SWITCHED OFF

Assigning one changes nothing until you edit it — each arrives with its one amount knob at zero, so A is dry and the first thing you reach for is B's amount. Eight of the ten are bit-exact at rest; TUBE's valve stage is in circuit at any drive, so that one colours the moment you assign it.

Two are deliberately not offered. SPACE rebuilds its reverb between blocks, so its character cannot travel. KIERANATOR is a drawn step pattern, and the grid that draws it is on the rack's face, not in a lane.

STEP BY STEP, AND IT WORKS

- 1 Insert it on the drum bus or the master. Nothing changes yet.
- 2 Lane 1: choose **CLIMB**. Click the lane. In the B row set CUTOFF to 300 Hz and RESO to 60 %. Until you do, the lane stays hatched. Leave ENTER at 0 and EXIT at 100 %.
- 3 Lane 2: **CHOP**. ENTER about 50 %, EXIT 95 %. A RATE 1/4, B RATE 1/16. CURVE = ACCEL, so the rate rushes at the end.
- 4 Lane 3: **RISER**. ENTER 30 %, EXIT 100 %. A LEVEL -60 dB, B LEVEL -6 dB, B FREQ 8000. SOURCE NOISE.
- 5 Lane 4: **GAP**. ENTER 92 %, EXIT 100 %. A DEPTH 0, B DEPTH 100. ON ARRIVAL = CLEAR.
- 6 Set **MONO GATE** to 60 %.
- 7 In the DAW draw a **POSITION** ramp 0→100 % over the last four bars. Draw an **ARRIVAL** blip that goes on just before the bar line of the drop, and back to off after.



Midway. The lanes the marker has passed carry heat; the ones ahead of it are still cold.

WHAT YOU WILL HEAR

The filter closes across four bars, the chop starts halfway and accelerates, the riser rises from bar two, the mix narrows in the last 15 %, the gap cuts the last half-beat, and on the bar the chain lets go.

Two more worth trying once that works: **DIVE** on the last bar with B SEMIS at -12, the whole track bending into the drop; and **BLOOM** with B FREEZE at 100 % and ON ARRIVAL = SPILL, so the wash is caught and hangs over the downbeat.

MEASURED, NOT ASSERTED

ENGINE BENCH — 506 CHECKS, ALL CLEAR

The score is obeyed, the loudness contract holds per effect, ARRIVAL is sample-accurate, rendering is identical at 64 and 256-sample blocks, and the six most expensive effects loaded at once cost about 10 % of one core.

ANTI-ALIASING

BRAKE and DIVE read a delay line faster than it was written, which folds everything above the new Nyquist back into the band. Both run on a 2× oversampled line. A 14 kHz tone folds at **−64.6 dB** for DIVE at +12 semitones, where it was 0.0 dB before; −64.6 for BRAKE launching out of a stop; and −103.3 for GRAIN at full SPREAD.

WRAPPER HARNESS — 57 CHECKS

Parameters, buses, state round trip, a rite from a newer build tolerated, scrubbing fires nothing, ARRIVAL fires, and a rite carrying World modules comes back as the same modules with the same knobs.

PANEL SNAPSHOT — 37 CHECKS

A travelling lane carries heat, a lane whose A equals its B carries none, and the lane menu is the eighteen with the ten World modules behind their own door, every one selectable.

NOT VERIFIED

Nobody has heard it in a DAW yet. Everything above is a machine reading numbers off real audio. None of it is the same as a person listening.

WHAT THE PANEL DOES NOT HAVE YET

Seven effects have no mark of their own. The glyph sheet carries eleven marks and the empty brackets; GAP falls back to its name in type. A World module in a lane has none either, and draws its name in the rack's teal instead.

The wordmark is not used. The delivered plank is shorter than its own lettering, so every letter is cut off at the bottom. The panel draws the title in type until that is redrawn.

Without a host clock the plug-in free-runs at 120 BPM, so ARRIVAL still lands on a bar of its own clock.



At rest. Every lane cold, the marker at zero.

LICENCE

AGPLv3. Read it, change it, build it. If you distribute a modified build, publish your changes. The source is in the same repository as this handbook, under `vst3-apps/rite-of-passage/plugin`. The artwork is not licensed for reuse.

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